

SimpleSwitch – Complete Documentation

1. Introduction

`SimpleSwitch` is a custom Flutter toggle switch widget designed as a lightweight alternative to the default Material `Switch`. It uses Flutter animations to provide smooth visual feedback and follows a **controlled widget** pattern.

This documentation explains: - What the widget does - How it works internally - How to use it correctly - The full source code

2. Widget Purpose

The purpose of `SimpleSwitch` is to: - Represent a boolean ON/OFF state - Allow parent widgets to control the state - Provide a clean, minimal UI with animation

The widget itself **does not store state**. Instead, it relies on the parent widget to pass the current value and update it when changes occur.

3. Public API

Constructor

```
const SimpleSwitch({  
  Key? key,  
  required bool value,  
  required ValueChanged<bool> onChanged,  
})
```

Parameters

Name	Type	Description
<code>value</code>	<code>bool</code>	Current switch state (<code>true</code> = ON, <code>false</code> = OFF)
<code>onChanged</code>	<code>ValueChanged<bool></code>	Callback triggered when the switch is tapped

4. Behavior Explanation

User Interaction

- The entire widget is wrapped in a `GestureDetector`
- When tapped, the switch toggles its value
- The new value is sent to the parent using `onChanged`

```
onTap: () => widget.onChanged(!widget.value)
```

Animation

- `AnimatedContainer` animates background color and border changes
- `AnimatedAlign` animates the knob movement
- Animation duration: **200 milliseconds**

5. Visual States

OFF State (`value = false`)

- Knob aligned to the left
- Background color: `AppColors.cBEBEBE`

ON State (`value = true`)

- Knob aligned to the right
- Background color: `AppColors.white`

6. Styling Details

Property	Value
Width	45
Height	18
Padding	1
Border Radius	14
Knob Size	20 x 20
Border Color	<code>AppColors.primary</code>
Knob Color	<code>AppColors.primary</code>

7. Full Source Code

```
import 'package:flutter/material.dart';
import 'package:nanduba/export.dart';

/// A custom animated toggle switch widget.
///
/// This is a controlled widget:
/// - The current state is provided using [value]
/// - State changes are returned via [onChanged]
///
/// The widget does not manage its own state.
class SimpleSwitch extends StatefulWidget {
  /// Indicates whether the switch is ON or OFF
  final bool value;

  /// Called when the user toggles the switch
  final ValueChanged<bool> onChanged;

  const SimpleSwitch({
    super.key,
    required this.value,
    required this.onChanged,
  });

  @override
  State<SimpleSwitch> createState() => _SimpleSwitchState();
}

class _SimpleSwitchState extends State<SimpleSwitch> {
  @override
  Widget build(BuildContext context) {
    return GestureDetector(
      onTap: () => widget.onChanged(!widget.value),
      child: AnimatedContainer(
        duration: const Duration(milliseconds: 200),
        width: 45,
        height: 18,
        padding: const EdgeInsets.all(1),
        decoration: BoxDecoration(
          border: Border.all(color: AppColors.primary),
          borderRadius: BorderRadius.circular(14),
          color: widget.value
            ? AppColors.white
            : AppColors.cBEBEBE,
        ),
      ),
    );
  }
}
```

```

    ),
    child: AnimatedAlign(
      duration: const Duration(milliseconds: 200),
      alignment: widget.value
        ? Alignment.centerRight
        : Alignment.centerLeft,
      child: Container(
        width: 20,
        height: 20,
        decoration: const BoxDecoration(
          shape: BoxShape.circle,
          color: AppColors.primary,
        ),
      ),
    ),
  ),
),
);
}
}

```

8. Usage Example

```

bool isEnabled = false;

SimpleSwitch(
  value: isEnabled,
  onChanged: (newValue) {
    setState(() {
      isEnabled = newValue;
    });
  },
);

```

9. Important Notes

- This widget must be rebuilt with a new value to update UI
 - Best used with `setState`, Provider, Riverpod, or Bloc
 - Can be easily extended with disabled state or custom sizes
-

10. Possible Enhancements

- Add `bool enabled` parameter
- Make colors configurable
- Convert to `StatelessWidget`
- Add accessibility semantics